MODULE 1: INTRODUCTION TO TECHNICAL COMMUNICATION

(CO: 2 | Weightage: 20 Marks in FAT — Questions 1 & 2)

Complete Study Textbook, Blueprint Guides, Writing Formats & Solved FAT Past Papers

Aligned with Official VIT BAENG101 Course Page Syllabus, School of Social Sciences & Languages Question Bank, and FAT Papers

Scope of This Module (Official Syllabus - All 6 Topics):
(1) The Nature of Technical Writing, (2) Taxonomy of Technical Writing: Scientific Writing vs. Technical Reporting, (3) Style Conventions and Audience Adaptations, (4) The 5 Cs of Technical Writing (Clarity, Conciseness, Completeness, Concreteness, Correctness), (5) Business Communication Principles & Channels, and (6) Professional Business Correspondence & Templates.

1 1. VIT FAT Exam Blueprint: Questions 1 & 2 Breakdown

In the VIT Final Assessment Test (100 Marks, 10 Questions × 10 Marks each), Module 1 directly accounts for the first **20 Marks** of your paper:

Question	Exam Focus	Detailed Scope & Evaluator Expectations	Marks
Question 1	Conceptual Technical Communication	Comparison: Scientific Writing vs. Technical Reporting, or General/Creative Writing vs. Technical Writing; Formal communication channels (downward, upward, lateral) and semantic/organizational barriers.	10 Marks
Question 2	The 5 Cs Applied Case Study	A poorly written, ambiguous, or wordy business memo/email is provided. You must: (1) Identify specific violations of the 5 Cs with quotes, and (2) Rewrite a perfected professional version using appropriate formats.	10 Marks

[VIT FAT EXAM INSIGHT] The 1 Trap in Question 2 (5 Cs Case Study):
Students frequently jump straight to rewriting the email without explicitly demonstrating their theoretical knowledge of the 5 Cs. This caps your score at 5 or 6 out of 10! To guarantee a full 10/10, your answer **MUST** contain two clearly titled sections:

1. **Section A: Diagnostic Critique Table** identifying which of the 5 Cs were violated in the prompt with supporting textual citations.
2. **Section B: Polished Professional Document** rewritten in full formal block format with complete metadata headers.

2 2. The Nature of Technical Writing & Core Attributes

Technical writing is transactional, purposeful, and reader-centered written communication designed to convey complex scientific, engineering, and managerial information clearly to enable effective decision-making or technical task execution.

Criterion	Technical Writing	Creative / General Writing
Primary Purpose	Utilitarian & transactional: to inform, instruct, record data, or persuade stakeholders to take action.	Aesthetic & expressive: to entertain, evoke emotion, or stimulate artistic imagination.

Target Audience	Specific, defined professional stakeholders (engineers, technicians, managers, regulatory clients).	Broad, general public with non-specific prior technical domain background.
Tone & Register	Objective, formal, impersonal, unambiguous, and third-person factual orientation.	Subjective, emotional, informal or conversational, frequently first-person.
Vocabulary	Precise domain-specific terminology, standardized SI units, operational definitions, zero ambiguity.	Figurative language, metaphors, idiomatic expressions, poetic hyperbole.
Visual Architecture	Highly structured: numbered headings, bullet points, data tables, flowcharts, schematics.	Continuous linear narrative paragraphs, dialogic pacing, minimal visual data display.

3 3. Taxonomy: Scientific Writing vs. Technical Reporting

A recurring 10-mark question in FAT papers (e.g. April 2025 Slot F2 Q1) requires contrasting **Scientific Writing** (Academic Journal Papers) and **Technical Reporting** (Industry Operational Reports).

Parameter	Scientific Writing (Academic Papers)	Technical Reporting (Industry Reports)
Primary Objective	To document fundamental original research, validate empirical hypotheses, and contribute to universal knowledge.	To solve immediate operational problems, evaluate commercial feasibility, and guide corporate decisions.
Target Audience	Specialized peer researchers, academics, journal reviewers, and scientific specialists.	Multi-disciplinary corporate stakeholders: C-suite executives, project managers, technicians.
Structural Model	Strictly follows IMRAD: Introduction, Methods, Results, and Discussion.	Executive Summary, Problem Statement, Methodology, Findings, Risk Matrix, Recommendations.
Tone & Citation	Impersonal academic passive voice; comprehensive literature citations (IEEE/ APA format).	Action-oriented, direct, concise; executive summaries, bulleted remediation steps; minimal literature citation.
Action Expected	Academic peer review, citation indexation, theoretical replication.	Immediate budgetary allocation, engineering modification, or safety risk remediation.

4 4. The 5 Cs of Technical Writing: Rules & Replacement Vault

The 5 Cs constitute the foundational quality benchmark for all professional engineering documentation:

4.1 4.1 Clarity (Freedom from Ambiguity)

- **Principle:** Write so that the message cannot possibly be misunderstood.
- **Eliminate Dangling Modifiers:**
  - **Incorrect:** "While inspecting the high-voltage transformer, a sudden electrical arc startled the technician." **(The arc was not inspecting the transformer!)**
  - **Correct:** "While inspecting the high-voltage transformer, the technician was startled by a sudden electrical arc."
- **Unambiguous Pronoun Reference:** Ensure pronouns like **it**, **they**, and **this** have an explicit antecedent noun.

4.2 4.2 Conciseness (Maximum Information Density)

- **Principle:** Prune redundant deadwood phrases without sacrificing essential meaning or courtesy.
- **Master Wordy Phrase Replacement Table:**

Wordy Flawed Phrase	Concise Form	Wordy Flawed Phrase	Concise Form
At this point in time	Now / currently	In the event that	If

Due to the fact that	Because	In order to	To
Has the capability of	Can	With reference to	Regarding / on
During the time that	While	For the purpose of	For / to
A large number of	Many	In close proximity to	Near
At an early date	Soon	Until such time as	Until
Prior to the start of	Before	In the majority of instances	Usually

4.3 4.3 Completeness (All Necessary Technical Parameters)

- Ensure all 5 Ws and 1 H (Who, What, Where, When, Why, and How) are answered.
- In engineering communications, never omit: Part numbers, exact model codes, testing temperatures, SI units, tolerances, and contact coordinates.

4.4 4.4 Concreteness (Specific Facts over Vague Generalizations)

- Replace subjective, hand-waving adjectives with empirical measurements:
  - **Vague:** "The electric motor ran extremely hot and slowed down considerably."
  - **Concrete:** "The electric motor's casing temperature reached 92 °C, reducing shaft speed from 1,800 RPM to 1,220 RPM."
  - **Vague:** "We need this shipment very quickly."
  - **Concrete:** "Delivery must be completed by 17:00 IST on Friday, November 13, 2026."

4.5 4.5 Correctness (Accuracy of Facts & Mechanics)

- Zero tolerance for factual data errors, misprinted serial numbers, grammatical disagreement (concord), or improper formal register.

5 5. Complete Professional Writing Formats & Templates

5.1 5.1 Format 1: Full-Block Business Letter Template

APEX DYNAMICS ENGINEERING PVT. LTD.
Plot 42, Electronics City Phase 1, Bengaluru, Karnataka — 560100

November 6, 2026

Dr. Arvind Sundaram
Director of Procurement, Titan Aviation Ltd.
Tech Park North, Chennai, Tamil Nadu — 600032

Subject: Formal Technical Inquiry regarding High-Precision Titanium Fasteners (RFQ 9042)

Dear Dr. Sundaram,

1. Opening (Context & Purpose): I am writing on behalf of Apex Dynamics Engineering to request technical specifications and commercial quotation terms for your Ti-6Al-4V Grade 5 aerospace fasteners for our upcoming defense airframe assembly line.

2. Body Paragraph (Technical Specifications & Parameters): Our engineering requirements demand fasteners capable of withstanding cyclic shear loads up to 45 kN at continuous operational temperatures of 320 °C. Specifically, we require delivery schedules for:

- 10,000 units of M8 × 45 mm Hex-head Titanium bolts (ISO 4014 standard)
- 10,000 units of matching self-locking nylon insert nuts (ISO 7040 standard)

3. Action Request & Closing: Please provide your product conformity certifications (AS9100D), unit pricing tiers, and estimated shipment lead times by November 16, 2026. Feel free to contact our senior mechanical lead, Er. Sneha Rao, at +91-80-2852-9000 for any technical clarifications.

Sincerely,

Kavya Narayanan
Senior Lead Procurement Engineer, Apex Dynamics Ltd.

5.2 5.2 Format 2: Formal Engineering Memorandum (Memo) Template

INTERNAL TECHNICAL MEMORANDUM

TO: All Departmental Project Engineers, Hardware Division
FROM: R. K. Verma, Principal Quality Assurance Lead
DATE: November 6, 2026
SUBJECT: Mandatory Static Voltage Discharge (ESD) Protocols for Cleanroom Assembly

1. Purpose & Background: During the third quarter inspection, 14% of assembled CMOS sensor boards exhibited gate dielectric breakdown traced directly to electrostatic discharge during manual handling.

2. Key Findings & Remediation Protocol: Effective immediately, all cleanroom assembly personnel must comply with the following ESD protective measures:
- Conductive wrist straps must be attached to verified ground terminals prior to touching component bins.
- Humidity in cleanrooms B1 and B2 will be held strictly between 45% and 55% RH to suppress static buildup.
- All non-conductive plastic storage containers must be replaced with static-dissipative conductive foam trays.

3. Implementation Timeline: Mandatory calibration audits will be conducted every morning at 08:30 IST. Non-compliant assembly batches will be quarantined.

6 6. Worked VIT FAT Past Papers & Model 10-Mark Questions

6.1 Problem 1.1 (BMAT101L / BAENG101 FAT April 2025 Slot F2 — 10 Marks)

Question: Differentiate between Scientific Writing and Technical Reporting with respect to: (i) Primary Objective, (ii) Intended Audience, (iii) Structural Organization, (iv) Tone and Style, and (v) Outcome/Action expected.

Model 10/10 Answer: Refer to the comprehensive 5-parameter comparative matrix in Section 3 of this textbook. When writing your exam answer:
1. State the formal definition of both genres in the opening paragraph.
2. Present the 5 parameters in a structured table.
3. Provide concrete examples: e.g. An IEEE Transactions journal paper on solid-state battery chemistry (Scientific Writing) vs. a 40-page feasibility audit on deploying battery swapping stations across Chennai metro corridors (Technical Reporting).

6.2 Problem 1.2 (FAT Question 2 Model Case Study: 5 Cs Analysis & Complete Rewrite — 10 Marks)

Question: Analyze the following poorly written email sent by an operations supervisor to a client. Identify the violations of the 5 Cs of technical writing, and rewrite the email in a professional business format: " *Hey guys, regarding that order you asked for, it is going to be late. The machines had some problems yesterday and stuff was broken so our workers couldn't finish it. We will try to get it done sometime next week probably. Let me know if that is okay or whatever.* "

Model 10/10 Answer:

Section A: Diagnostic Critique (Identification of 5 Cs Violations)

Violated "C"	Textual Evidence from Prompt	Diagnostic Analysis & Correction
Clarity	"regarding that order… stuff was broken"	Fails to cite the specific Purchase Order number, part description, or technical nature of equipment downtime.
Conciseness	"try to get it done sometime next week probably… or whatever"	Uses informal fillers, sloppy phrasing, and colloquial wordiness.
Completeness	"going to be late… workers couldn't finish"	Omits crucial parameters: specific units delayed, revised ship date, courier tracking, and contingency options.

Concreteness	"sometime next week probably"	Replaces actionable firm calendar dates with vague, non-committal estimates.
Correctness	"Hey guys… if that is okay or whatever"	Inappropriate informal slang for a formal contractual B2B transaction; unprofessional tone.

Section B: Rewritten Professional Business Email

Subject: Schedule Update: Purchase Order 48102 (Precision CNC Machined Shafts)

Dear Mr. Henderson,

I am writing to provide an immediate status update regarding Purchase Order 48102 for 500 units of hardened steel drive shafts, originally scheduled for dispatch on Friday, November 6, 2026.

Due to an unexpected hydraulic pump failure on CNC Milling Station 3 yesterday afternoon, machining operations were suspended for 8 hours for emergency seal replacement. Station 3 has been recalibrated and resumed full production at 09:00 IST this morning.

To mitigate delivery impact on your assembly schedule, we have expedited production shifts and arranged split shipments:

- **Batch 1 (250 Units):** Dispatched via BlueDart Air Cargo on Monday, November 9, 2026 (Delivery: Nov 10).
- **Batch 2 (250 Units):** Dispatched on Wednesday, November 11, 2026 (Delivery: Nov 12).

All air freight surcharges will be absorbed entirely by our firm. We deeply regret this operational delay and remain committed to supporting your project timeline. Please contact me directly at +91-44-2250-9800 for any further assistance.

Sincerely,
Arunachalam Murugan
Operations Manager, Precision Components Ltd.

7 7. Quick Self-Audit Checklist for Exam Day

1. Did you answer Question 2 with **BOTH** a 5 Cs critique table and a full professional rewrite?
2. Did you replace all vague words (**“very fast”**, **“soon”**) with numerical benchmarks (**“within 48 hours”**, **“at 1500 RPM”**)?
3. Did you verify that your business letter includes date, inside address, explicit subject line, and formal complimentary close?